Harsh Amin

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Education

University of British Columbia
BSc Co-op Computer Science · Math *Vancouver, BC*
GPA: 4.00

Awards: Trek Academic Excellence Scholarship - awarded to top 5% of faculty
Extracurricular: UBC Launch Pad, Intramural Hockey, Weight Lifting Club, Entrepreneurship Club

Experience

Machine Learning Engineer Intern – AI Center of Excellence May 2026 – Aug 2026
Amazon *Vancouver, BC*

- Cut agent-test authoring time by **85%** by building a test-generation platform (**AWS CDK** construct) that turns live production traffic into approval-gated regression tests
- Eliminated ∼**95%** of redundant test cases by building a trace-clustering pipeline using **BGE-M3** embeddings and an ensemble density-based clustering approach, distilling **8,000+** production agent executions into ∼**380** distinct behavioural scenarios across **2** structurally different agents (a RAG assistant and a sequential tool-use agent)
- Cut the regression-escape rate by **40%** by designing a behavioural-drift detector that auto-generates new simulations when novel production behaviour emerges, gated by a review dashboard on **CloudFront** and **Fargate**
- Improved behavioural-cluster coherence by **20%+** by introducing a sequence-aware distance metric (**Dynamic Time Warping**) that groups agent runs by execution order rather than tool set alone

GenAI Software Developer Intern May 2025 – Dec 2025
UBC Cloud Innovation Centre (Collab. w/Amazon Web Services) *Vancouver, BC*

- Shipped **3+** open-source **GenAI** apps end-to-end for sponsors, incl. a BCcampus study platform for **100,000+** students running a **LangChain RAG** pipeline over **750+** textbooks on **AWS Bedrock**
- Designed and deployed an enterprise-grade **AI Agent** using a **Model Context Protocol (MCP)** server; integrated **DynamoDB** for continuous conversational memory, sustaining context across deep multi-turn sessions and improving query precision by **30%**
- Migrated from unreliable local Docker workflows to a standardized **AWS CI/CD** pipeline (**CodeBuild, CodePipeline, ECR**), cutting cross-environment inconsistencies by **33%** and reducing deployment time by **40%**
- Codified a reusable **AWS CDK** security baseline (3-tier **VPC**, private endpoints, **RDS Proxy** in isolated subnets, tiered **WAF** rate limits) as **Claude Code** and **Kiro** agent skills, reused for every new app

President (previously Technical Lead & Developer) Sep 2024 – Present
UBC Launch Pad – UBC's largest student software engineering team *Vancouver, BC*

- Lead **50+** members in **5+** product teams building software for UBC labs, non-profits and local companies
- Raised **$2,000+** in sponsorships and run hackathons and tech mixers for UBC's builder community
- As tech lead, rebuilt Forum AI's **RAG** pipeline on **FastAPI** and **PGVector**, cutting API response time by **27%**

Publications and Projects

SEGRA: A Structured Experience Guided Reasoning Agent | *EMNLP & COLM Workshop* Jul 2026

- Co-authored **SEGRA**, an experience-guided agent that answers enterprise questions by generating **Gremlin** graph queries over knowledge graphs
- Demonstrated a **7.0x** higher mean judge score than baseline **chain-of-thought prompting** while reducing **LLM** calls by **20%** and inference costs by **18%**

FlowGuard | *Winner, Rogers × Databricks Hackathon* Sep 2026

- Built a TransLink operations console that forecasts crowd exits up to **2 h** ahead from anonymized Rogers cell-tower data, reaching $\mathbf{R^2}$ **0.93–0.99** on unseen months vs. **0.64** for a typical-week baseline
- Shipped end-to-end on **Databricks**: a Unity Catalog medallion pipeline on serverless Jobs, **MLflow**-tracked models, and a **Genie** agent for natural-language operator queries

Technical Skills

Languages: Python, Java, C/C++, SQL, TypeScript/JavaScript, R, GraphQL
Machine Learning: PyTorch, scikit-learn, Hugging Face, NumPy, pandas, Embeddings, Clustering, LLM Evaluation
LLMs & Agents: LangChain, LlamaIndex, Agents, Ollama, AWS Bedrock, RAG, Model Context Protocol (MCP)
MLOps & Data: Databricks, MLflow, SageMaker, PostgreSQL/PGVector, Vector Databases, MongoDB, Docker
Cloud & Tools: AWS (CDK, Lambda, Fargate), FastAPI, Flask, React, Node.js, Git, Linux/Unix, Jenkins
AI-Native Dev: Claude Code, Codex, Kiro, opencode, MCP Servers, Agent Skills & Hooks, LLM-as-Judge Evals